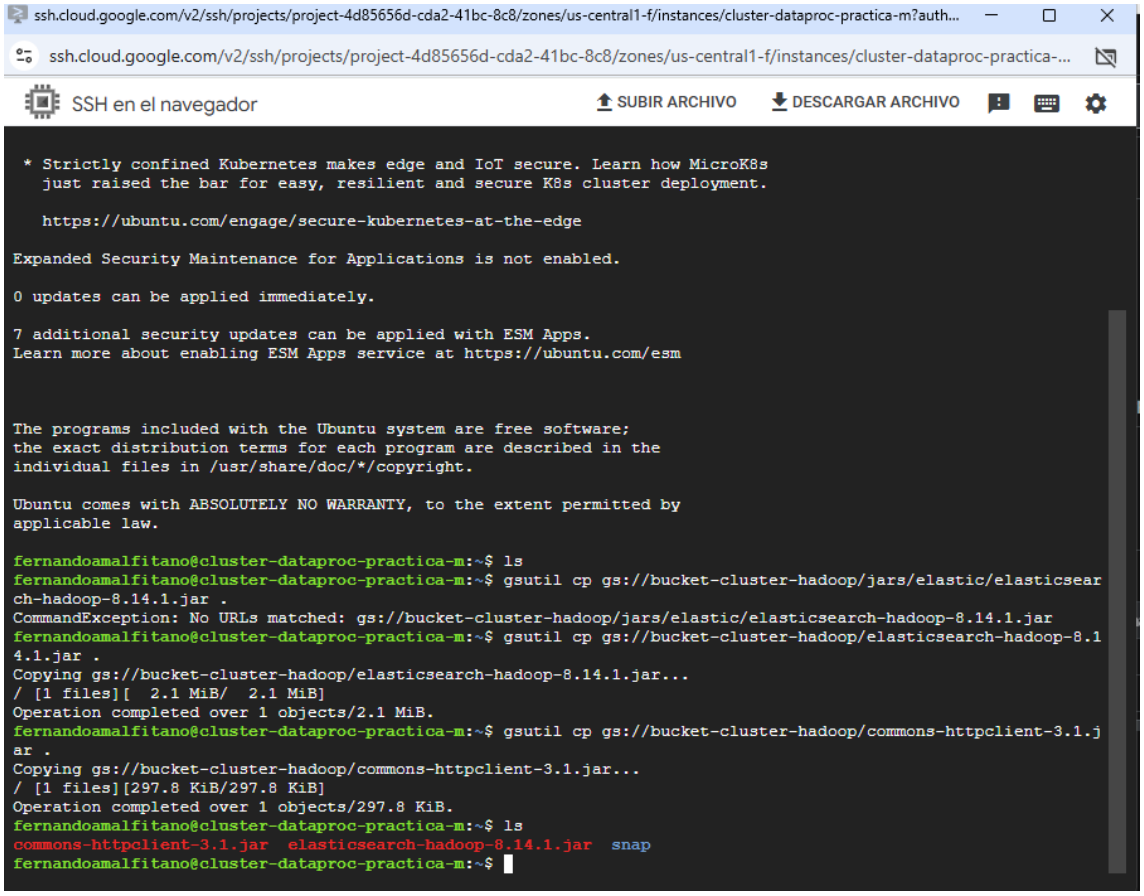
PRÁCTICA ARQUITECTURA BIG DATA

Fernando Amalfitano

PARTE 1:



PARTE 2:

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/elastic-search-practica?authuser=...

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/elastic-search-practica?aut...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
#action.destructive_requires_name: false

#----------------------- BEGIN SECURITY AUTO CONFIGURATION -----------------------
#
# The following settings, TLS certificates, and keys have been automatically
# generated to configure Elasticsearch security features on 16-03-2026 18:56:15
#
# ----------------------------------------------------------------------------

# Enable security features
xpack.security.enabled: false

xpack.security.enrollment.enabled: true

# Enable encryption for HTTP API client connections, such as Kibana, Logstash, and Agents
xpack.security.http.ssl:
  enabled: false
  keystore.path: certs/http.p12

# Enable encryption and mutual authentication between cluster nodes
xpack.security.transport.ssl:
  enabled: true
  verification_mode: certificate
  keystore.path: certs/transport.p12
  truststore.path: certs/transport.p12
# Create a new cluster with the current node only
# Additional nodes can still join the cluster later
cluster.initial_master_nodes: ["elastic-search-practica"]

# Allow HTTP API connections from anywhere
# Connections are encrypted and require user authentication
http.host: 0.0.0.0

# Allow other nodes to join the cluster from anywhere
# Connections are encrypted and mutually authenticated
#transport.host: 0.0.0.0

#----------------------- END SECURITY AUTO CONFIGURATION -------------------------
fernandoamalfitano@elastic-search-practica:~$
```

No es seguro 34.132.240.40:9200

BOOTCAMP KeepCodingBDML17 KC-Modulos-transv... Go

Dar formato al texto


```
{
  "name" : "elasticsearch-practica-1",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "A12G-pL7S42T1bXOwCdpeA",
  "version" : {
    "number" : "8.14.1",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "93a57a1a76f556d8aee6a90d1a95b06187501310",
    "build_date" : "2024-06-10T23:35:17.114581191Z",
    "build_snapshot" : false,
    "lucene_version" : "9.10.0",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

PARTE 3:

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/cluster-dataproc-practica-m?auth...

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/cluster-dataproc-practica-...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

fernandoamalfitano@cluster-dataproc-practica-m:~$ ls
fernandoamalfitano@cluster-dataproc-practica-m:~$ gsutil cp gs://bucket-cluster-hadoop/jars/elastic/elasticsear
ch-hadoop-8.14.1.jar .
CommandException: No URLs matched: gs://bucket-cluster-hadoop/jars/elastic/elasticsearch-hadoop-8.14.1.jar
fernandoamalfitano@cluster-dataproc-practica-m:~$ gsutil cp gs://bucket-cluster-hadoop/elasticsearch-hadoop-8.1
4.1.jar .
Copying gs://bucket-cluster-hadoop/elasticsearch-hadoop-8.14.1.jar...
/ [1 files][  2.1 MiB/  2.1 MiB]
Operation completed over 1 objects/2.1 MiB.
fernandoamalfitano@cluster-dataproc-practica-m:~$ gsutil cp gs://bucket-cluster-hadoop/commons-httpclient-3.1.j
ar .
Copying gs://bucket-cluster-hadoop/commons-httpclient-3.1.jar...
/ [1 files][297.8 KiB/297.8 KiB]
Operation completed over 1 objects/297.8 KiB.
fernandoamalfitano@cluster-dataproc-practica-m:~$ ls
commons-httpclient-3.1.jar  elasticsearch-hadoop-8.14.1.jar  snap
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo sed -i '$d' /etc/hive/conf.dist/hive-site.xml
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>es.nodes</name>\n
 <value>35.239.41.251</value>\n  </property>\n' /etc/hive/conf.dist/hive-site.xml
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>es.port</name>\n
<value>9200</value>\n  </property>\n' /etc/hive/conf.dist/hive-site.xml
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>es.nodes.wan.only</n
ame>\n    <value>true</value>\n  </property>\n' /etc/hive/conf.dist/hive-site.xml
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo sed -i '$a \  <property>\n    <name>hive.aux.jars.path</
name>\n   <value>/usr/lib/hive/lib/elasticsearch-hadoop-8.14.1.jar,/usr/lib/hive/lib/commons-httpclient-3.1.jar
</value>\n  </property>\n</configuration>' /etc/hive/conf.dist/hive-site.xml
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo cp elasticsearch-hadoop-8.14.1.jar /usr/lib/hive/lib/
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo cp commons-httpclient-3.1.jar /usr/lib/hive/lib/
fernandoamalfitano@cluster-dataproc-practica-m:~$ sudo systemctl restart hive-server2
fernandoamalfitano@cluster-dataproc-practica-m:~$
```

PARTE 4:

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/cluster-dataproc-practica-m?auth...

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/cluster-dataproc-practica-...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO


```
{"errors":false,"took":98,"items":[{"index":{"_index":"alumnos","_id":"3","_version":1,"result":"created","_sha
rds":{"total":2,"successful":1,"failed":0},"_seq_no":1,"_primary_term":1,"status":201}},{"index":{"_index":"alu
mnos","_id":"4","_version":1,"result":"created","_shards":{"total":2,"successful":1,"failed":0},"_seq_no":2,"_p
rimary_term":1,"status":201}},{"index":{"_index":"alumnos","_id":"5","_version":1,"result":"created","_shards":
{"total":2,"successful":1,"failed":0},"_seq_no":3,"_primary_term":1,"status":201}},{"index":{"_index":"alumnos"
,"_id":"7","_version":1,"result":"created","_shards":{"total":2,"successful":1,"failed":0},"_seq_no":4,"_primar
y_term":1,"status":201}},{"index":{"_index":"alumnos","_id":"8","_version":1,"result":"created","_shards":{"tot
fernandoamalfitano@cluster-dataproc-practica-m:~$ curl -X GET "http://35.239.41.251:9200/alumnos/_search?pretty
"oc-practica-m:~$ curl -X GET "http://35.239.41.251:9200/alumnos/_search?pretty"
{
  "took" : 23,
  "timed_out" : false,
  "_shards" : {
    "total" : 1,
    "successful" : 1,
    "skipped" : 0,
    "failed" : 0
  },
  "hits" : {
    "total" : {
      "value" : 6,
      "relation" : "eq"
    },
    "max_score" : 1.0,
    "hits" : [
      {
        "_index" : "alumnos",
        "_id" : "6",
        "_score" : 1.0,
        "_source" : {
          "title" : "New Document",
          "content" : "This is a new document for the master class",
          "tag" : [
            "general",
            "testing"
          ]
        }
      },
      {
```

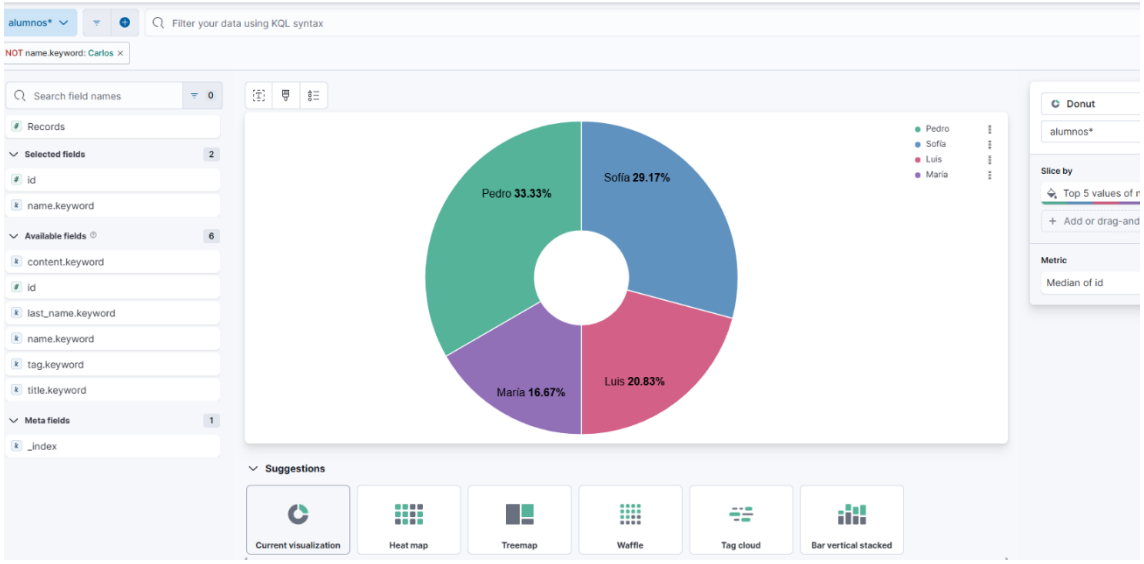

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/cluster-dataproc-practica-m?auth...

ssh.cloud.google.com/v2/ssh/projects/project-4d85656d-cda2-41bc-8c8/zones/us-central1-f/instances/cluster-dataproc-practica-...

SSH en el navegador SUBIR ARCHIVO DESCARGAR ARCHIVO


```
        "id" : 4,
        "name" : "María",
        "last_name" : "López"
      }
    },
    {
      "_index" : "alumnos",
      "_id" : "5",
      "_score" : 1.0,
      "_source" : {
        "id" : 5,
        "name" : "Luis",
        "last_name" : "Martínez"
      }
    },
    {
      "_index" : "alumnos",
      "_id" : "7",
      "_score" : 1.0,
      "_source" : {
        "id" : 7,
        "name" : "Sofía",
        "last_name" : "Ramírez"
      }
    },
    {
      "_index" : "alumnos",
      "_id" : "8",
      "_score" : 1.0,
      "_source" : {
        "id" : 8,
        "name" : "Pedro",
        "last_name" : "Hernández"
      }
    }
  ]
}
}
fernandoamalfitano@cluster-dataproc-practica-m:~$
```

PARTE 5:



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